

Specification of Experimental Therapeutic Device

Introduction

This document describes the concept of an experimental device inspired by Tumor Treating Fields (TTF) therapy. The goal is to create a wearable device that generates an oscillating electric field in the **200–300 kHz** range using a copper induction coil (“pancake coil”) and a system of concentric iron rings (“Russian doll matrix”) placed at the front of the torso. This is intended only as a prototype for research purposes.

Main Components

1. Iron Matrix (Russian Doll rings):

- Material: ferromagnetic steel (laminated or solid).
- Geometry: concentric circular rings.
- Example dimensions:
 - Inner ring radius: ≈ 10 cm, thickness 5 mm.
 - Middle ring radius: ≈ 15 cm, thickness 5 mm.
 - Outer ring radius: ≈ 20 cm, thickness 5 mm.
 - Central disk (optional): circular steel plate, radius ≈ 8 –10 cm.
- Number of elements: 3 rings + central disk.
- Purpose: redirect and shape the electric field inward toward the body.

2. Copper Coil (Inductor):

- Type: flat spiral (“pancake”) coil.
- Wire: insulated copper, diameter 1–1.5 mm.
- Turns: 10–20, single layer.
- Outer diameter: ≈ 25 cm (to cover the outer ring).
- Connection: two ends terminated with a jack/connector for generator link.

3. Signal Generator:

- Frequency: adjustable in the range 150–350 kHz.
- Output: sufficient amplitude to induce an internal field of about 1–3 V/cm (to be determined experimentally).

- Components: oscillator (DDS or analog), MOSFET driver stage, power amplifier feeding the coil.
- Control: potentiometer for power regulation, basic indicators (on/off, thermal limit).
- Safety: current limit, thermal cutoff at 70°C (coil) or 42°C (skin side).

4. Power Supply (Battery):

- Type: lithium-ion, portable.
- Capacity: 12 V, 10–15 Ah (approx. 150–180 Wh).
- Usage: 3 batteries per day (12+ hours of operation with rotation).
- Connection: via DC jack to generator.

Mounting and Attachment

- The matrix and coil are integrated into a front capsule (housing).
- Attachment to the body via Velcro straps or textile harness.
- The generator and battery may be carried in a small backpack on the back.

Expected Operation

- Continuous or pulsed mode (25–50% duty cycle) to limit heating.
- Temperature monitoring: NTC sensors and/or bimetallic thermostats.
- Therapeutically useful effect expected when internal field ≥ 1 V/cm in tissue.

Conclusion

The described device is a prototype system combining an induction coil and an iron matrix to shape an oscillating field in biological tissue. This document is intended for engineers to guide the construction of an initial model and does not represent a finished medical device.